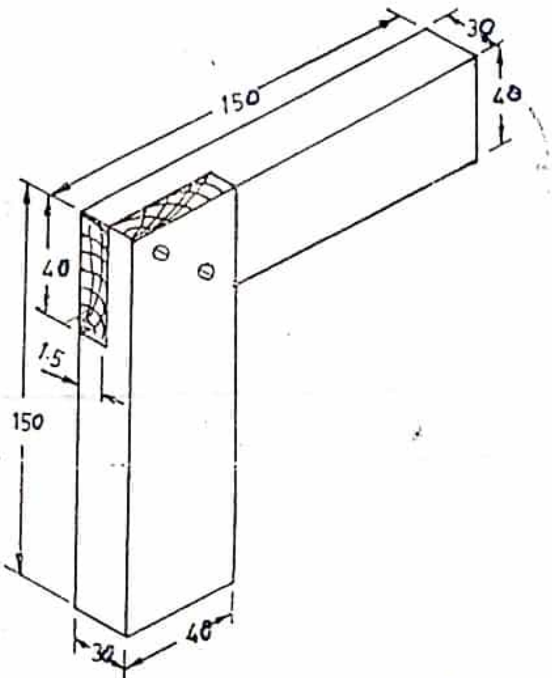
CARPENTRY SHOP

PRACTIÇE-I

Time : 3 Hrs.

OBJECT:- To make a corner half lap joint, as per given drawing.

MATERIAL REQUIRED :- Mango wood size 35mm x 45mm x 305mm, wooden screw 25mm.



ALL DIMENSIONS ARE IN mm.

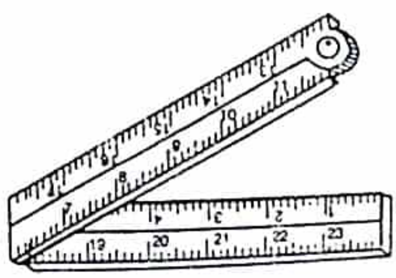
TOOLS AND EQUIPMENTS USED:-

1) **Measuring and Marking Tools:**
   - a) Folding rule
   - b) Try Square
   - c) Scriber
   - d) Marking Gauge
2) **Holding and supporting Tools:**
   - a) Work Bench
   - b) Carpenter Bench vice
3) **Cutting Tools:**
   - a) Iron Jack Plane
   - b) Tenon Saw
   - c) Hand Drill
4) **Striking Tools:** Not required
5) **Miscellaneous Tools:**
   - a) Screw Driver

DESCRIPTION OF TOOLS AND EQUIPMENTS:-

1. a) **Folding Rule :**

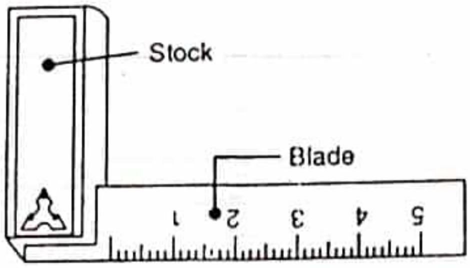
It is a wooden scale consisting of four pieces each 150mm long joined together by hinges. It is mainly used for measuring and setting out dimensions.



Four-fold rule.

b) Try Square :

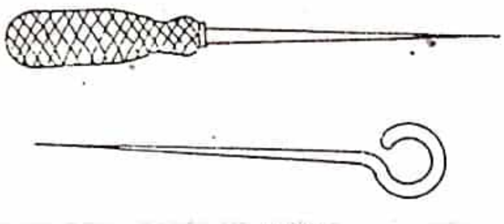
It consist of a steel blade fitted into a metallic stock at a right angle. It is used for measuring and testing of the planed surfaces, draw parallel line at right angle to a planed surface, draw mutually perpendicular lines over a plane surface and test the squareness of two adjacent surfaces .



Try square.

c) Scriber:

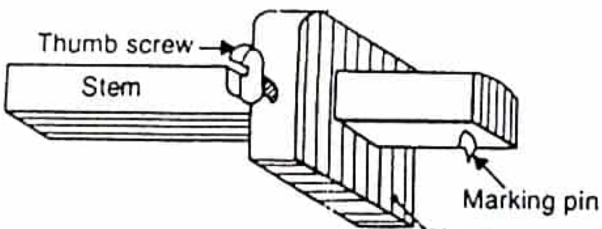
It is a steel rod having sharp point at one end. It is mainly used for locating and marking points and scribing lines on wood surface.



Scriber

d) Marking Gauge :

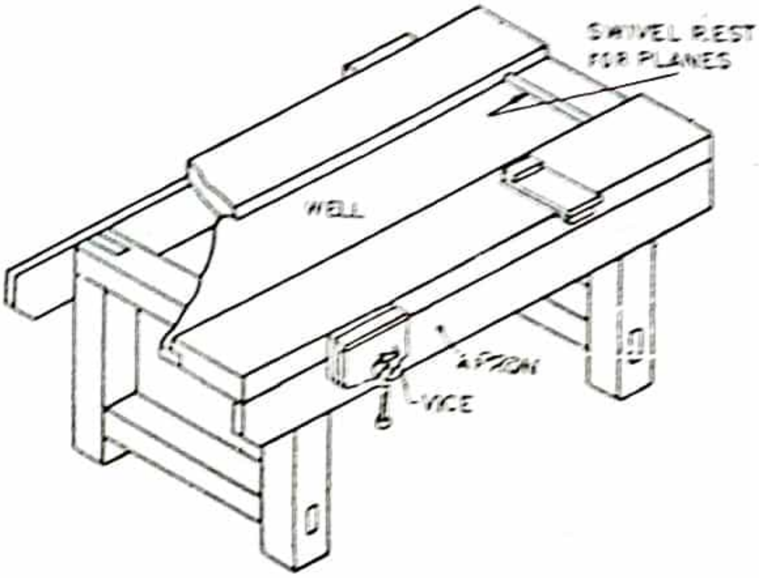
It is consisting of wooden stem stock of rectangular or square cross-section, and scribing pin as shown in figure. It is used for scribing lines parallel to a finish face or edge.



Marking gauge.

2. a) Work Bench :

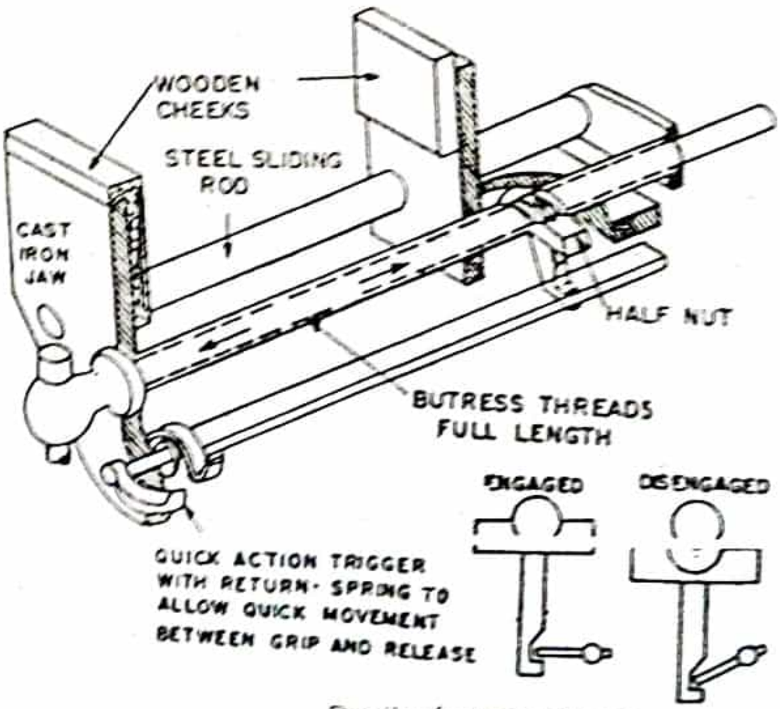
It is heavy table of rigid construction made of hard wood two or four carpenters vice are fitted on opposite side to hold the work piece during the operations.



carpenter's work bench.

b) Carpenters Bench Vice :

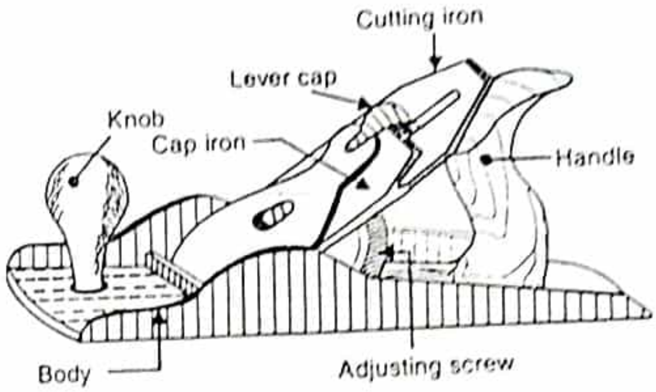
Vice are fitted on work bench to hold the job during operations inside opposite faces of the jaws are fitted with wooden liners so as to prevent damaging of job surfaces.



Details of carpenter's vice.

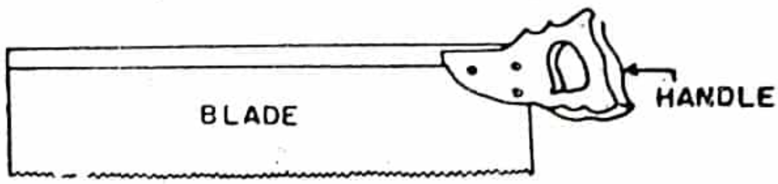
3. a) **Iron Jack Plane :**

It is used for planing of wood surfaces. Its whole body is made of cast iron provided with wooden handle all the main parts of this plane shown in fig.



b) **Tenon Saw:**

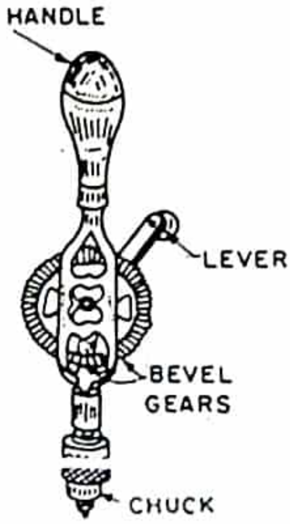
It has a parallel blade, 25 to 45cm long and 6 to 10cm wide having 5 to 8 teeth per cm. length the main use of this saw is in taking short straight cut such as for tenon for this reason its blade is provided with back at the top so that the blade does not bend during the operation.



Tenon saw.

c) **Hand Drill :**

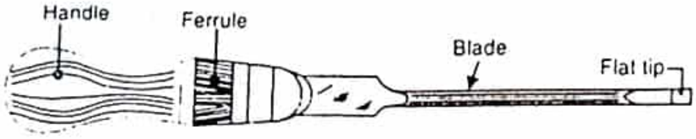
Its consist of forged body on the top of which is provided wooden handle and bottom a chuck as shown in fig. it is used for drilling small holes only.



Hand drill.

5. a) **Screw Driver :**

It is used for driving the wood screws. It consist of a wooden or plastic handle and a steel blade.



Screw Driver

PROCEDURE :-

1. Measure the work piece size.
2. Hold the work piece in vice.
3. Plane one broad surface of the work piece and check the trueness using a try square.
4. Plane remaining work piece surfaces with respect to True surface and check its trueness and dimensions.
5. Mark the middle point of the length and mark a line each on two adjacent surface to divide the job into two equal parts.
6. Hold the job again and cut i along the marked lines by means of the tenon saw. Thus, Two identical wooden piece of 30mm x 40mm x 150mm size each are obtained.
7. Set the half thickness of work piece in marking gauge. On both the pieces mark from one end of the job upto 40mm length as shown in object figure with the help of scale, try square scriber and marking gauge.
8. By means of Tenon saw cut the marked line along its vertical faces of both pieces.
9. Remove the unwanted wood from this part.
10. Drilling on one work piece for wooden screw by means of hand drill.
11. After finishing both pieces in the above manner check their dimensions and finish one again.
12. After ensuring the accuracy of dimensions of both the parts separately fit them together with screw with the help of screw driver to obtain the required joint.

PRECAUTIONS :-

1. Cutting should be done away from the body during sawing.
2. Tools are always kept well on working bench.
3. The wooden work piece should be hold rigidly on the bench vice-before operation.
4. Tools should operated by handling from the right position.
5. Always wear shoes and apron during operation.

PRACTICE-II

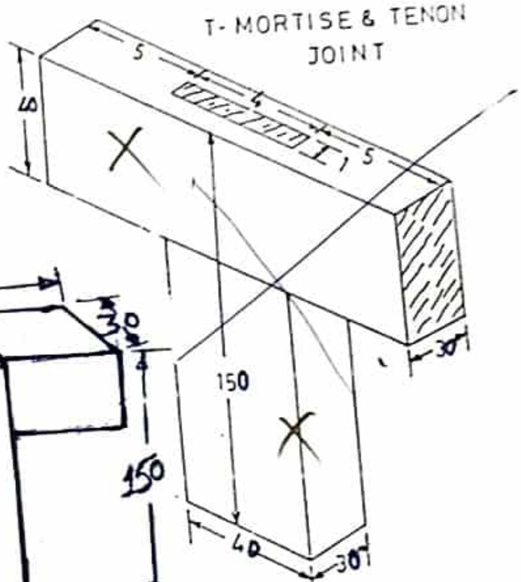
Time : 3 Hrs.

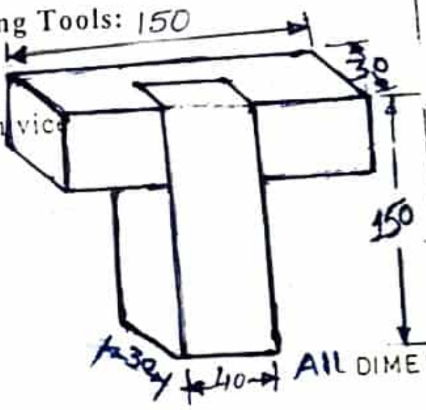
OBJECT:- To make a mortise and Tenon Joint as per given drawing

MATERIAL REQUIRED :- Mango wood size 35mm x 45mm x 305mm.

TOOLS AND EQUIPMENTS USED:-

1) **Measuring and Marking Tools:**
   - a) Folding rule
   - b) Try Square
   - c) Scriber
   - Marking d) ~~Mortise~~ Gauge
2) **Holding and supporting Tools:**
   - a) Work Bench
   - b) Carpenter Bench vice
3) **Cutting Tools:**
   - a) Iron Jack Plane
   - b) Tenon Saw
   - c) ~~Mortise Chisel~~
   - d) Firmer Chisel
4) **Striking Tools:**
   - a) Wooden Mallet





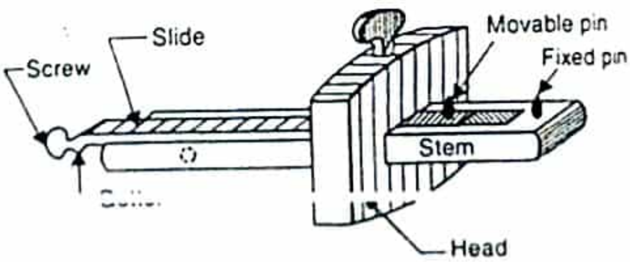
All DIMENSION ARE IN mm.

T Half Joint

DESCRIPTION OF TOOLS AND EQUIPMENTS :-

a) **Mortise Gauge :**

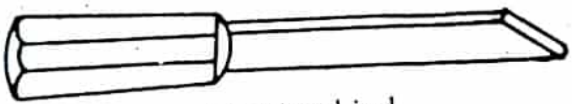
It is improved form of marking gauge, it carries a significant feature that instead of only one scribing pin it has two pins as shown in the figure. One of the pin is movable while the other is fixed.



Mortise gauge.

b) Mortise Chisel :

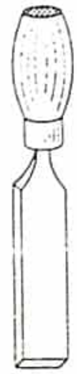
It is used for taking heavy and deep cuts resulting more stock removal because of more thickness its blade is stronger than other chisels of same size and is therefore, capable of being subjected to heavy loads. A typical form of mortise chisel is shown in figure.



Mortise chisel.

c) Firmer Chisel :

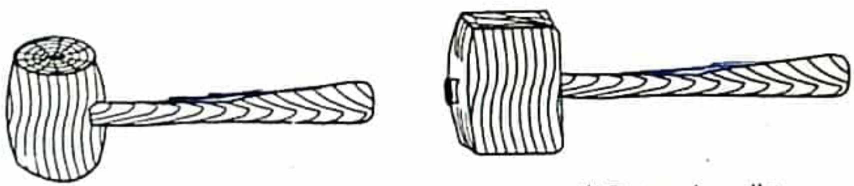
It is general purpose carpentry chisel. It carries a wide blade of rectangular cross section. It is used for taking wider cuts and finish flat surfaces inside the grooves.



Firmer Chisel

d) Wooden Mallet :

It is made of hard wood and is rectangular or round in shape. Provided with wooden handle. It is used for striking the cutting tools.



(*a*) Round mallet. (*b*) Rectangular mallet.

Types of mallets.

PROCEDURE :-

1. Measure the work piece size.
2. Hold the work piece in vice and plane one broad surface and check the trueness using a try square.

3. Repeat the operation if required and make this surface true.
4. Then plane the adjacent narrow surface and check its trueness and also its squareness with the previously planed surface.
5. Similarly, plane the remaining two surface and check the size & squareness of the all surfaces.
6. Cut this piece into two halves after marking a midway line on two equal piece.
7. Perform marking using mortise gauge on both the pieces according to the dimensions.
8. Using Tenon saw cut away the unwanted material from one piece.
9. Dig out the unwanted wood from another piece with help of mortise chisel.
10. Clean up the wider internal surface of mortise with the help of firmer chisel.
11. Then the fit the two pieces together to obtain the joint.

PRECAUTIONS :-

1. Cutting should be done away from the body during sawing.
2. Tools are always kept well on working bench.
3. The wooden work piece should be hold rigidly on the bench vice-before operation.
4. Tools should operated by handling from the right position.
5. Always wear shoes and apron during operation.